

Question	Marking Guidance	Mark	Comments
08.1	1. Increasing IAA concentration increases (cell wall) plasticity OR IAA activates enzymes which increases (cell wall) plasticity; 2. Increase in (cell wall) plasticity results in <u>cell</u> elongation;	2 (2 x AO2)	1 and 2. Accept 'stretching' for plasticity, Reject 'elasticity' once only. 2. Idea of <u>cell</u> elongation or increase in <u>cell</u> length must be conveyed. 2. IAA causing cell elongation on its own is not enough.

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08.2	Correct answer of 570 (%) = 2 marks;; (However, if working shows plasticity values used, e.g. 8 and 1.2 (or difference of 6.8) and answer of 570 (%) = 1 mark) Answer or working shows 560 / 566 / 567 (%) = 1 mark (However, if working shows plasticity values used, e.g. 8 and 1.2 (or difference of 6.8) and answer of 560 / 566 / 567 (%) = 0 marks) Answer of 120 (%) = 1 mark OR Answer of 230 (%) = 1 mark OR Answer of 340 (%) = 1 mark OR Answer of 780 (%) = 1 mark OR Answer of 1200 (%) = 1 mark;	2 (2 x AO2)	Accept 566 recurring for one mark. Ignore any numbers after 560 or 566 or 567.

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08.3	1 part/cm³ of 10⁻¹/stock/GA and/to 9 parts/cm³ of (distilled) water, then 1 part/cm³ (of 10⁻²) and/to 9 parts/cm³ of (distilled) water OR	1 (1 x AO2)	Accept any volumes equivalent in parts to example provided.

	0.1 parts/cm ³ of 10 ⁻¹ /stock/GA and/to 9.9 parts/cm ³ of (distilled) water;		Accept '1 part/cm ³ of 10 ⁻¹ /stock/GA and/to 9 parts/cm ³ of (distilled) water, then repeat'.
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08.4	Max 3 marks from mark points 3 to 8. 1. (Use distilled) water (control) and different (GA) concentrations and 10 (stem) segments in each; 2. Measure (length of) stem/segments at start and at end (of investigation) OR Determine increase/change in length OR Determine percentage change in length; 3. (Place stems in same) volume of solution/GA; 4. (Leave for same period of) time; 5. (Same) temperature; 6. (Same) species/type (of plant); 7. (Same) age (of plant); 8. (Same) diameter/thickness of stem/segments;	5 max (5 x AO2)	1. Accept use of specified number of segments in each dish with repeats, if total used = 60. 2. Accept 'cut to same length' = measure at start. 2. Accept 'growth in length' as 'increase/change in length'. 3 to 8. Ignore pH, carbon dioxide, humidity, nutrients, and light. 4. Accept any 'named period of time'. 6. Accept 'same plant'. 7. Accept (stems) at same stage of growth/development.

